

PHYS-467 Machine Learning for Physicists
Graded Homework n. 1: Linear Regression
Due: October 29, 2021

Please, upload your solution to the Moodle page of the course. No delay is accepted.

Linear regression and the role of regularization

1. Download the data-set from the moodle: week 11-17 October, folder HW 1. Split it into three sets: one for training, one for validation, and one for testing.
2. Plot the learning curve (i.e. test error vs number of training samples) of the least squares linear regression with no regularization. Consider number of training samples $n > d$ up to the full size of your training set. Comment on the behaviour of the error.
3. Add a square (l_2 norm) regularization to the least square loss with $\lambda = 1$ and with $\lambda = 0.001$ and re-plot the learning curve, this time starting from $n = 10$. Comment.
4. Now vary the regularization strength λ and plot the components of the optimal weights as a function of λ . What is the effect of increasing λ ? Consider two different training set sizes n , namely $n =$ your full training set size and $n = 30$. Using the validation set, compute the error as a function of λ and find the optimal value λ^* for both values of n . How does λ^* change? Print the optimal weights. Finally, evaluate on the test set for both values of n .
5. Consider now the l_1 -norm regularization. Plot the regularization path, i.e. the weights as a function of varying λ (this will also allow you to see what are relevant ranges of λ , they may be small or large, you may need to search a bit). Comment on what is the effect of this type of regularization. Find the optimal λ^* for $n =$ your full training set size and $n = 30$, print the optimal weights, and evaluate the final model on the test set.
6. Which of the regularizations performs better for this data-set? How does this depend on the training set size? Comment.